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Electronic apparatus

Abstract

According to an embodiment, an electronic apparatus includes a printed circuit board including a plurality of devices that include a nonvolatile memory package and a controller package configured to control the nonvolatile memory package, and a housing accommodating the printed circuit board. The housing includes an opening on a surface constituting the housing. An encryption device among the plurality of devices is present in a first region. The first region is a region on the printed circuit board that is not irradiated with light emitted from a light source placed at the opening. The encryption device is a device used for an encryption process of data to be stored into the nonvolatile memory package.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5563450	12/1995	Bader	N/A	N/A
6460772	12/2001	Stahl	N/A	N/A
8879263	12/2013	Gunderson	N/A	N/A
9078357	12/2014	Gunderson	N/A	N/A
2007/0274032	12/2006	Ni	361/679.31	H05K 5/0269
2011/0038123	12/2010	Janik	N/A	N/A
2011/0199748	12/2010	Kagawa	361/796	H05K 7/20436
2014/0085818	12/2013	Lee	N/A	N/A
2017/0060199	12/2016	Kim	N/A	N/A
2017/0220293	12/2016	Kim	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-173375	12/1997	JP	N/A
2001-111262	12/2000	JP	N/A
2002-110896	12/2001	JP	N/A
2009-170760	12/2008	JP	N/A
2011-170566	12/2010	JP	N/A
2014-032655	12/2013	JP	N/A
2015-201984	12/2014	JP	N/A

OTHER PUBLICATIONS

“Implementation Guidance for FIPS PUB 140-2 and the Cryptographic Module Validation Program”, National Institute of Standards and Technology Communications Security

Establishment, http://csrc.nist.gov/groups/STM/cmvp/documents/fips140-2/FIPS1402IG.pdf, 2017, 216 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 17/660,075 filed Apr. 21, 2022, which is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 17/120,982 filed Dec. 14, 2020, which is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 16/662,558, filed Oct. 24, 2019, which is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 15/824,125, filed Nov. 28, 2017, which is based upon and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2017-031040, filed Feb. 22, 2017, and based upon and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2017-149384, filed Aug. 1, 2017, the entire contents of each of which are incorporated herein by reference.

FIELD
(1) Embodiments described herein relate generally to an electronic apparatus.
BACKGROUND
(2) An electronic apparatus is known in which a housing accommodates boards with packages mounted thereon and the packages contain a nonvolatile semiconductor memory and a controller for controlling the nonvolatile memory. In this electronic apparatus, a base or cover constituting the housing is provided with openings for attaching and detaching connectors.
(3) In the case of the conventional structure, the packages or the like mounted on the boards can be visually observed through the openings. On the other hand, there is a demand that the production number or model number of a package, such as a package for constituting a controller, should be protected from visual observation. However, according to the conventional structure of the electronic apparatus, it is difficult to satisfy this demand.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating an example of an appearance configuration of an electronic apparatus according to a first embodiment;
(2) FIG. **2** is a sectional view illustrating an example of an internal configuration of the electronic apparatus according to the first embodiment;
(3) FIG. **3** is an exploded perspective view illustrating an example of the electronic apparatus according to the first embodiment;
(4) FIG. **4** is an exploded perspective view illustrating an example of a board assembly according to the first embodiment;
(5) FIG. **5** is a sectional view schematically illustrating an example of the arrangement position of a controller package according to the first embodiment;
(6) FIG. **6** is a side view schematically illustrating an example of the relation between the positions of ventilation holes and the position of the board assembly, according to the first embodiment;

(7) FIG. **7** is a side view schematically illustrating an example of the relation between the positions of ventilation holes and the position of the board assembly, according to the first embodiment;
(8) FIG. **8** is a side view schematically illustrating an example of the relation between the positions of ventilation holes and the position of the board assembly, according to a comparative example;
(9) FIGS. **9**A and **9**B are sectional views illustrating an example of an internal configuration of an electronic apparatus according to a second embodiment;
(10) FIG. **10** is an exploded perspective view illustrating an example of the electronic apparatus according to the second embodiment;
(11) FIG. **11** is an exploded perspective view illustrating an example of a board assembly according to the second embodiment;
(12) FIGS. **12**A and **12**B are perspective views illustrating an example of frames according to the second embodiment;
(13) FIGS. **13**A to **13**C are top views illustrating examples of a ventilation mechanism of a frame according to the second embodiment;
(14) FIG. **14** is a perspective view illustrating another example of the ventilation mechanism of a frame according to the second embodiment;
(15) FIGS. **15**A to **15**E are views illustrating an example of an assembling sequence of the board assembly according to the second embodiment;
(16) FIGS. **16**A and **16**B are views illustrating an example of the cover of a housing according to a third embodiment;
(17) FIG. **17** is a sectional view illustrating another example of the cover of the housing according to the third embodiment;
(18) FIG. **18** is a perspective view illustrating an example of a board assembly according to the third embodiment; and
(19) FIG. **19** is a perspective view illustrating a configuration example of a spacer according to the third embodiment.

DETAILED DESCRIPTION

(20) According to one embodiment, an electronic apparatus includes a printed circuit board including a plurality of devices that include a nonvolatile memory package and a controller package configured to control the nonvolatile memory package, and a housing accommodating the printed circuit board. The housing includes an opening on a surface constituting the housing. An encryption device among the plurality of devices is present in a first region. The first region is a region on the printed circuit board that is not irradiated with light emitted from a light source placed at the opening. The encryption device is a device used for an encryption process of data to be stored into the nonvolatile memory package.
(21) Exemplary embodiments of an electronic apparatus will be explained below in detail with reference to the accompanying drawings. The present invention is not limited to the following embodiments.

First Embodiment

(22) FIG. **1** is a perspective view illustrating an example of an appearance configuration of an electronic apparatus according to a first embodiment. FIG. **2** is a sectional view illustrating an example of an internal configuration of the electronic apparatus according to the first embodiment. FIG. **3** is an exploded perspective view illustrating an example of the electronic apparatus according to the first embodiment. FIG. **4** is an exploded perspective view illustrating an example of a board assembly according to the first embodiment. Hereinafter, the electronic apparatus will be exemplified by a Solid State Drive (SSD) that uses a nonvolatile memory as a storage medium. Further, hereinafter, for the sake of convenience, it is assumed that the latitudinal direction of the rectangular top surface or bottom surface of the electronic apparatus is an X-direction, its longitudinal direction is a Y-direction, and its thickness direction is a Z-direction. Furthermore, hereinafter, the relative positional relations of components arranged in the Z-direction, i.e., their

relations in the vertical direction, will be illustrated, by using the arrangement state of the electronic apparatus illustrated in FIGS. **1** and **2** as a reference.
(23) The electronic apparatus **1** has a flat rectangular parallelepiped shape as its physical appearance. The electronic apparatus **1** includes a housing **10** having a hollow rectangular parallelepiped shape, and a board assembly **20** including one or more boards accommodated in the housing **10**.
(24) The housing **10** includes a base **11** and a cover **12**. The base **11** includes a bottom wall **111** like a flat plate, and side walls **112***a* and **112***b* perpendicularly extending upward in the Z-direction from the outer periphery of the bottom wall **111**. In this example, the base **11** is provided with a pair of side walls **112***a* each having a surface perpendicular to the X-direction, and a single side wall **112***b* arranged at one end in the Y-direction and having a surface perpendicular to the Y-direction.
(25) The side walls **112***a* are provided with screw holes **113** extending in the Z-direction. The thickness of the side walls **112***a* is set larger at the portions formed with the screw holes **113** than at the other portions. In this example, the screw holes **113** are arranged near the both ends and near the center in the Y-direction.
(26) Further, the side walls **112***a* are provided with a plurality of pins **114**. The pins **114** are disposed on the side walls **112***a* and projected upward in the Z-direction (i.e., toward the cover **12**). In this example, the pins **114** are arranged at two places near the screw holes **113** provided on the side walls **112***a*. The pins **114** are provided to perform positioning in the X-direction and Y-direction to the cover **12** with respect to the base **11** at the time of placing the cover **12** onto the base **11**. Accordingly, the cover **12** is provided with through holes **125** for inserting the pins **114** therein, at the corresponding positions.
(27) The bottom wall **111** of the base **11** is provided with screw holes **115**. The screw holes **115** are provided to fix the board assembly **20** to the base **11** by fasteners, such as screws **142**. Further, the bottom wall **111** of the base **11** is provided with a pin (not illustrated). The pin is disposed on the bottom wall **111** and projected upward in the Z-direction. The pin may be disposed on a pedestal (not illustrated) provided on the bottom wall **111**. The pin is provided to perform positioning in the X-direction and Y-direction to the board assembly **20** with respect to the base **11**. Accordingly, a printed circuit board **21**A to be arranged at the lowermost side is provided with a through hole **213** for inserting this pin therein, at the corresponding position.
(28) The cover **12** includes a top wall **121** like a flat plate, and side walls **122** perpendicularly extending downward in the Z-direction from the outer periphery of the top wall **121**. In this example, the cover **12** is provided with a pair of side walls **122** each having a surface perpendicular to the Y-direction. Further, the side walls **122** are provided with ventilation holes **123** for cooling, which allow air to flow between the outside and inside of the housing **10**.
(29) Heat conductive sheets **131** serving as heat conductive members are provided at predetermined positions on the bottom wall **111** of the base **11** and the top wall **121** of the cover **12**. Each heat conductive sheet **131** is made of silicone resin, for example, and has thermally conductive and electrically insulating properties, as well as an elastic property. Each heat conductive sheet **131** is provided to conduct heat generated by a device on the board assembly **20** to the base **11** or cover **12**, to suppress a temperature rise inside the housing **10**. Accordingly, the heat conductive sheets **131** are arranged to be in contact with devices on the upper surface of the board assembly **20** and the top wall **121** of the cover **12**, and to be in contact with devices on the lower surface of the board assembly **20** and the bottom wall **111** of the base **11**.
(30) The cover **12** is provided with through holes **124** for inserting screws **141** therein, and the through holes **125** for inserting the pins **114** therein. The through holes **124** for inserting the screws **141** are arranged at the positions corresponding to the screw holes **113** of the base **11**. The through holes **125** for inserting the pins **114** are arranged at the positions corresponding to the pins **114** of the base **11**. Further, a nameplate label **151** with information, such as the model number of the electronic apparatus **1**, described therein is stuck to the upper surface of the cover **12**.

(31) The base **11** and the cover **12**, which constitute the housing **10**, are made of aluminum die cast or the like superior in heat radiation property.
(32) The board assembly **20** includes one or more printed circuit boards as the boards. When the board assembly **20** includes one printed circuit board, the printed circuit board is fixed to the housing **10** by fasteners. Further, when the board assembly **20** includes a plurality of printed circuit boards, the board assembly **20** further includes one or more spacers, and the printed circuit boards and the spacers are alternately stacked in the Z-direction, and are fixed by fasteners. In this example, the board assembly **20** has a structure in which three printed circuit boards **21**A, **21**B, and **21**C and two spacers **22**A and **22**B are alternately stacked in the Z-direction.
(33) On the respective printed circuit boards **21**A to **21**C, devices are mounted. The devices are circuit components, which include nonvolatile memory packages **23***a*, volatile memory packages **23***b*, a controller package **23***c*, and capacitors **23***d*. Each nonvolatile memory package **23***a* has a configuration such that a nonvolatile semiconductor memory chip using a NAND type flash memory, for example, is packaged by heat resistant resin, ceramics, or the like. Each volatile memory package **23***b* has a configuration such that a Dynamic Random Access Memory (DRAM) chip or a Static RAM (SRAM) chip is packaged by heat resistant resin, ceramics, or the like. The controller package **23***c* has a configuration such that a controller chip for controlling the nonvolatile memory packages **23***a* and the volatile memory packages **23***b* is packaged by heat resistant resin, ceramics, or the like. The controller package **23***c* is formed of a System-on-a-Chip (SoC), for example. Each capacitor **23***d* plays a role to assist the supply of power supplied from a host apparatus to which the electronic apparatus **1** is connected.
(34) The controller package **23***c* serves to control the exchange of data with the host apparatus. Specifically, upon receiving a data write command from the host apparatus, the controller package **23***c* temporarily stores data to be written, into a write buffer provided in the volatile memory packages **23***b*, and writes the data stored in the write buffer into the position corresponding to a specified address inside the nonvolatile memory packages **23***a*. Further, upon receiving a data read command from the host apparatus, the controller package **23***c* reads data from the position corresponding to a specified address inside the nonvolatile memory packages **23***a*, and temporarily stores the read data into a read buffer provided in the volatile memory packages **23***b*. Then, the controller package **23***c* sends the data stored in the read buffer to the host apparatus.
(35) Further, the controller package **23***c* includes a function of encrypting data to be written at the time of writing the data into the nonvolatile memory packages **23***a*, and a function of decrypting the encrypted data at the time of reading the data from the nonvolatile memory packages **23***a*. When the controller package **23***c* performs the encryption process and decrypting process of this kind (which will be simply referred to as “encryption process”) by itself, the controller package **23***c* is considered as an encryption module (encryption device). On the other hand, when another package is used in addition to the controller package **23***c* to perform the encryption process, such that, for example, data to be treated by the encryption process is read into the volatile memory packages **23***b* to perform the encryption process, each of the packages associated with the encryption process is considered as the encryption module. Alternatively, when a dedicated hardware, such as a dedicated circuit, is used to perform the encryption process, without using the controller package **23***c*, the dedicated hardware is considered as the encryption module.
(36) Each device is mounted on at least one of the two main surfaces of each of the printed circuit boards **21**A to **21**C by, for example, surface mounting or the like. In this example, the controller package **23***c* is arranged on the lower surface of the lowermost printed circuit board **21**A, and the volatile memory packages **23***b* and the capacitors **23***d* are arranged on the upper surface of the lowermost printed circuit board **21**A. Further, the nonvolatile memory packages **23***a* are arranged on the upper surface and lower surface of each of the printed circuit boards **21**A to **21**C.
(37) In addition to the devices, the printed circuit boards **21**A to **21**C are provided with board-to-board connectors **24** that mutually and electrically connect the individual printed circuit boards

stacked in the Z-direction. The board-to-board connectors **24** are mounted on the respective printed circuit boards **21**A to **21**C by surface mounting. The board-to-board connectors **24** are arranged at positions that are opposite to each other when the printed circuit boards **21**A to **21**C are superposed in an aligned state. The board-to-board connectors **24** include a female type connector **24***a* as one type and a male type connector **24***b* as the other type, which can be mutually fitted to achieve electrical connection between the printed circuit boards **21**A to **21**C. Thus, when the printed circuit boards **21**A to **21**C are mutually connected by the board-to-board connectors **24**, the printed circuit boards **21**A to **21**C are relatively firmly connected to each other.

(38) The printed circuit board **21**A arranged at the lowermost side is provided with the female type connector **24***a* on the upper surface. The printed circuit board **21**C arranged at the uppermost side is provided with the male type connector **24***b* on the lower surface. The printed circuit board **21**B arranged at the middle is provided with the male type connector **24***b* on the lower surface, and the female type connector **24***a* on the upper surface. Further, the lowermost printed circuit board **21**A is provided with a connector **25** at one end in the Y-direction, such that the connector **25** is to be electrically connected to the host apparatus present outside. As the standard of the connector **25**, for example, Peripheral Component Interconnect express (PCIe™) or Serial Attached Small Computer System Interface (SAS) is used.

(39) The printed circuit boards **21**A to **21**C are provided with through holes **212** for inserting the screws **142** that fix the board assembly **20** to the base **11**. These through holes **212** are arranged corresponding to the screw holes **115** of the bottom wall **111** of the base **11**. Further, the printed circuit board **21**A arranged at the lowermost side is provided with the through hole **213** corresponding to the pin provided on the bottom wall **111** of the base **11**. Further, the printed circuit boards **21**A and **21**B are provided with through holes **214** corresponding to pins **226** provided on the spacers **22**A and **22**B described later.

(40) Each of the spacers **22**A and **22**B is interposed between two of the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction. Each of the spacers **22**A and **22**B holds a state where two of the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction have a predetermined interval therebetween. Further, each of the spacers **22**A and **22**B includes a function of making it difficult for devices arranged between two of the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction to be visually observed, through the ventilation holes **123**. Each of the spacers **22**A and **22**B includes a frame part **221**, which has almost the same dimensions as those of the printed circuit boards **21**A to **21**C and has almost the same outline as that of the printed circuit boards **21**A to **21**C, and a reinforcing part **222** or reinforcing parts **222***a* and **222***b*, which suppress deformation of the frame part **221** caused by an external force. The frame part **221** is provided with a plurality of engaging portions **223** of a snap fit type at predetermined positions. The plurality of engaging portions **223** are preferably arranged opposite to each other. In the example illustrated in FIG. **4**, the engaging portions **223** are arranged on respective ones of a pair of long sides of the frame part **221**, and are opposite to each other.

(41) Each engaging portion **223** includes a cantilever projected downward from the frame part **221** at a predetermined position, and a protrusion provided at the distal end of the cantilever and projected toward the inside of the frame part **221**. On the other hand, the printed circuit boards **21**A and **21**B are provided with cutouts **211** at the positions corresponding to the engaging portions **223**, such that each cutout **211** is recessed by the thickness of the cantilever.

(42) The frame part **221** is provided with through holes **225** for inserting the screws that fix the board assembly **20** to the base **11**. Further, the frame part **221** is provided with a plurality of pins **226** projected downward. The pins **226** are used to perform positioning in the X-direction and Y-direction with respect to each of the printed circuit boards **21**A and **21**B to be fitted. Accordingly, the printed circuit boards **21**A and **21**B are provided with through holes **214** for fitting the pins **226** therein, at the corresponding positions.

(43) In assembling, first, the printed circuit board **21**A or **21**B is put closer to the lower side of the

spacer **22**A or **22**B. Then, while the positions of the through holes **214** of the printed circuit board **21**A or **21**B are set aligned with the positions of the pins **226** of the spacer **22**A or **22**B, the printed circuit board **21**A or **21**B is brought into contact with the spacer **22**A or **22**B. This results in a state where the protrusions of the engaging portions **223** are positioned on the lower surface side of the printed circuit board **21**A or **21**B, and engage with this lower surface. In this way, the printed circuit boards **21**A and **21**B are fixed to the spacers **22**A and **22**B.
(44) The reinforcing parts **222**, **222***a*, and **222***b* are arranged at any positions. For example, in the upper side spacer **22**B in FIG. **4**, the reinforcing part **222** is provided by connecting the opposite long sides almost in parallel with the short sides constituting the frame part **221**. Further, in the lower side spacer **22**A in FIG. **4**, the first reinforcing part **222***a* is provided by connecting the opposite long sides almost in parallel with the short sides constituting the frame part **221**, and the second reinforcing part **222***b* is provided by connecting the first reinforcing part **222***a* to one of a pair of the short sides almost in parallel with the long sides constituting the frame part **221**. The second reinforcing part **222***b* is arranged to extend along the backside of the arrangement position of the controller package **23***c*.
(45) The second reinforcing part **222***b* is provided with a raised portion **224** projected downward. In this example, the raised portion **224** is arranged within the arrangement position of the controller package **23***c* on the lowermost printed circuit board **21**A. Between the raised portion **224** and the lowermost printed circuit board **21**A, a heat conductive sheet **131** is interposed. The raised portion **224** has a height to come into contact with the heat conductive sheet **131** provided on the upper surface of the lowermost printed circuit board **21**A. Consequently, heat generated by an operation of the controller package **23***c* mounted on the lower surface of the printed circuit board **21**A is conducted to the upper surface of the printed circuit board **21**A, and is further conducted to the spacer **22**A through the heat conductive sheet **131**. Then, the heat is released by air flows coming from the ventilation holes **123**, i.e., openings. Further, as described above, the controller package **23***c* on the lower surface of the printed circuit board **21**A is connected to the bottom wall **111** of the base **11** through a heat conductive sheet **131**. Accordingly, heat generated by the controller package **23***c* is conducted also to the base **11** through the heat conductive sheet **131**, and is released.
(46) Further, in this example, between a portion of the second reinforcing part **222***b* which is above the arrangement position of the raised portion **224** and the nonvolatile memory packages **23***a* which is on the lower surface of the printed circuit board **21**B, a heat conductive sheet **131** is interposed in contact with the both sides. Consequently, heat generated by the nonvolatile memory packages **23***a* is conducted to the spacer **22**A through the heat conductive sheet **131**, and is released by air flows coming from the ventilation holes **123**.
(47) The dimensions of the spacers **22**A and **22**B in the Z-direction are set such that devices mounted on the printed circuit boards **21**A to **21**C to be arranged above and below the spacers **22**A and **22**B do not interfere with each other.
(48) The spacers **22**A and **22**B are made of an electrically insulating material that will not be thermally deformed by a temperature rise due to an operation of the controller package **23***c* and so forth, and that is high in resistance to impact shock and high in thermally conductive property. For example, the spacers **22**A and **22**B are made of polycarbonate resin.
(49) Here, in the above example, each of the reinforcing parts **222** and **222***a* is provided by connecting a pair of opposite sides of the frame part **221**; however, the embodiment is not limited to this. For example, each of the reinforcing parts **222** and **222***a* may be provided by connecting two sides of the frame part **221** that are not opposite to each other.
(50) Next, with reference to FIGS. **3** and **4**, an explanation will be given of a method of assembling such an electronic apparatus **1**. First, as illustrated in FIG. **4**, the board assembly **20** is assembled with the printed circuit boards **21**A to **21**C and the spacers **22**A and **22**B. Specifically, while the through holes **214** of the printed circuit boards **21**A and **21**B and the pins **226** of the spacers **22**A and **22**B are set aligned with each other, the printed circuit boards **21**A and **21**B and the spacers

22A and **22**B are put closer to each other in the Z-direction, and are fixed by the engaging portions **223** of the spacers **22**A and **22**B. In the illustrated example, the lowermost printed circuit board **21**A and the spacer **22**A are fixed to each other, and the middle printed circuit board **21**B and the spacer **22**B are fixed to each other. Then, the printed circuit boards **21**A to **21**C adjacent in the Z-direction are connected to each other by the board-to-board connectors **24**. Consequently, the board assembly **20** is constructed in which the printed circuit boards **21**A to **21**C and the spacers **22**A and **22**B are alternately arranged in the Z-direction.

(51) Then, as illustrated in FIG. **3**, the electronic apparatus **1** is assembled. First, the heat conductive sheets **131** are stuck to the base **11** and the cover **12** at predetermined positions. Thereafter, while the through hole **213** of the lowermost printed circuit board **21**A constituting the board assembly **20** is set aligned with the pin (not illustrated)of the base **11**, the board assembly **20** is placed on the base **11**. Then, the board assembly **20** is fixed to the base **11** by fasteners, such as the screws **142**. Specifically, in this fixing, the screws **142** are set to pass through the through holes **212** of the printed circuit boards **21**A to **21**C constituting the board assembly **20** and the screw-insertion through holes **225** of the spacers **22**A and **22**B, and to reach the screw holes **115** of the base **11**.

(52) Thereafter, while the through holes **125** of the cover **12** are set aligned with the pins **114** provided on the side walls **112***a* of the base **11**, the cover **12** is placed on the base **11**. Then, the cover **12** is fixed to the base **11** by fasteners, such as the screws **141**. Specifically, in this fixing, the screws **141** are set to pass through the through holes **124** provided on the cover **12**, and to reach the screw holes **113** provided on the side walls of the base **11**. Then, the nameplate label **151** is stuck to the upper surface of the cover **12**, and thereby the electronic apparatus **1** is assembled.

(53) Next, an explanation will be given of the ventilation holes **123** provided on the electronic apparatus **1** that accommodates the board assembly **20**. FIG. **5** is a sectional view schematically illustrating an example of the arrangement position of the controller package according to the first embodiment. For the sake of convenience in explanation, FIG. **5** illustrates a sectional view including one end in the Y-direction at a position where the connector **25** is not present. In FIG. **5**, the constituent elements corresponding to those described above are denoted by the same reference symbols.

(54) The housing **10** includes a side wall **122** formed with the ventilation holes **123** and arranged at one end in Y-direction. Now, it is assumed that, at the ventilation holes **123** of this side walls **122**, artificial light sources **100** are placed to irradiate light (visible light) of which a wavelength is 400 nm to 750 nm. In FIG. **5**, optical paths at outer edges of light emitted from each ventilation hole **123** are schematically illustrated by arrows, and the range of light emitted from each ventilation hole **123** is schematically illustrated by hatching of lines oblique downward to the left. This hatching is shown between the optical paths at two outer edges of light emitted from each ventilation hole **123**.

(55) In this case, an encryption module, i.e., the controller package **23***c* in this example, is arranged at a position where the light does not reach. For example, when it is assumed that, of the upper surface and lower surface of the board assembly **20**, a light unreachable region is a first region and a light reachable region is a second region, the controller package **23***c* is arranged in the first region. Here, the upper surface and lower surface of the board assembly **20** within the range illustrated in FIG. **5** are the first region.

(56) As described above, as the spacers **22**A and **22**B are interposed between the printed circuit boards **21**A to **21**C stacked in the Z-direction, devices arranged in the spaces between the printed circuit boards **21**A to **21**C cannot be visually observed through the ventilation holes **123**. Further, as the controller package **23***c*, which is considered as the encryption module, is present in the first region, the encryption module (controller package **23***c*) is prevented from being visually observed through the ventilation holes **123**. Thus, in relation to the encryption module (for example, the controller package **23***c*), it is possible to prevent the production number, model number, design

information (such as wire trace and internal structure), or assembling information from being visually observed through the ventilation holes **123**.
(57) Here, an explanation of the arrangement positions of the ventilation holes **123** in the Z-direction relative to the board assembly **20**, which are set to prevent the encryption module from being visually observed, will be given. In the following examples, there are a case (A) where the encryption modules are arranged on both of the upper and lower surfaces of the board assembly **20**; a case (B) where the encryption module is arranged on one of the upper and lower surfaces of the board assembly **20**; and a case (C) where no encryption module is arranged on either of the upper and lower surfaces of the board assembly **20**. Further, in the following examples, the encryption modules include the controller package **23***c* and the nonvolatile memory packages **23***a*.
(58) In the case (A) where the encryption modules are arranged on both of the upper and lower surfaces of the board assembly **20**:
(59) In this case, as illustrated in FIG. **5**, within an arrangement position range R.sub.0 of the board assembly **20** in the Z-direction, the ventilation holes **123** provided on the housing **10** are arranged within a range R.sub.1, which is between the upper surface and lower surface of the board assembly **20**. This is to prevent light from reaching the upper surface and lower surface of the board assembly **20** when the artificial light sources **100** are placed at the ventilation holes **123**. Consequently, the upper surface and lower surface of the board assembly **20** become the light unreachable first region. Thus, the nonvolatile memory packages **23***a* on the upper surface of the board assembly **20** and the controller package **23***c* on the lower surface of the board assembly **20** are arranged in the light unreachable first region.
(60) In the case (B) where the encryption module is arranged on one of the upper and lower surfaces of the board assembly **20**:
(61) FIG. **6** is a side view schematically illustrating an example of the relation between the positions of the ventilation holes and the position of the board assembly, according to the first embodiment. FIG. **6** illustrates a case where the encryption module is arranged on the lower surface of the board assembly **20**, for example. In this case, the ventilation holes **123** provided on the housing **10** are arranged within a range R.sub.1 between the lower surface of the board assembly **20** and the top wall **121** of the cover **12**. Accordingly, as long as this condition is satisfied, the ventilation holes **123** may be arranged also above the uppermost position of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction.
(62) In the example of FIG. **6**, the upper surface of the board assembly **20** is the second region to be irradiated with light, and the lower surface of the board assembly **20** is the first region not to be irradiated with light. In this case, as the encryption module is not arranged on the upper surface side of the board assembly **20**, the encryption module is prevented from being visually observed even if the upper surface of the board assembly **20** is irradiated with light from the artificial light sources **100** placed at the ventilation holes **123**. On the other hand, as the ventilation holes **123** are not arranged below the lowermost position of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction, the lower surface of the board assembly **20** will never be irradiated with light from the artificial light sources **100** placed at the ventilation holes **123**.
(63) Here, FIG. **6** illustrates a case where the encryption module is arranged on the lower surface of the board assembly **20**. On the other hand, where the encryption module is arranged on the upper surface of the board assembly **20**, the positional relations described above in the vertical direction become reverse to those illustrated in FIG. **6**.
(64) Further, if an alteration of firmware for controlling an operation of the electronic apparatus **1** can cause a problem in security, the encryption module including, for example, the nonvolatile memory packages **23***a* that store the firmware is preferably arranged at a position not to be visually observed.
(65) In the case (C) where no encryption module is arranged on either of the upper and lower surfaces of the board assembly **20**:

(66) FIG. **7** is a side view schematically illustrating an example of the relation between the positions of ventilation holes and the position of the board assembly, according to the first embodiment. FIG. **7** illustrates a case where no encryption module is arranged on either of the upper and lower surfaces of the board assembly **20**, for example. In this case, the ventilation holes **123** provided on the housing **10** are arranged at any positions within a range R.sub.1, which is between the top wall **121** of the cover **12** and the bottom wall **111** of the base **11**, regardless of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction. In other words, the ventilation holes **123** may be arranged also above the uppermost position of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction, or the ventilation holes **123** may be arranged also below the lowermost position of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction.
(67) In the example of FIG. **7**, the upper surface and lower surface of the board assembly **20** become the second region to be irradiated with light. In this case, as the encryption module is not arranged on either of the upper surface side and lower surface side of the board assembly **20**, the encryption module is prevented from being visually observed even if the upper surface and lower surface of the board assembly **20** are irradiated with light from the artificial light sources **100** placed at the ventilation holes **123**.
(68) Further, in this case, the encryption module results in being arranged in a region surrounded by the spacers **22**A and **22**B of the board assembly **20**. In the region surrounded by the spacers **22**A and **22**B, the encryption module will never be irradiated with light from the artificial light sources **100** placed at the ventilation holes **123**. In the example of FIG. **7**, the controller package **23***c* is arranged on the upper surface of the lowermost printed circuit board **21**A.
(69) Further, if an alteration of firmware for controlling an operation of the electronic apparatus **1** can cause a problem in security, the encryption module including, for example, the nonvolatile memory packages **23***a* that store the firmware is preferably arranged at a position not to be visually observed.
(70) As a comparative example, an explanation will be given of the arrangement positions of the ventilation holes **123** in the Z-direction relative to the board assembly **20**, in a case where the encryption modules are arranged on both of the upper and lower surfaces of the board assembly **20**, and the encryption modules can be visually observed through the ventilation holes **123**. FIG. **8** is a side view schematically illustrating an example of the relation between the positions of the ventilation holes and the position of the board assembly, according to the comparative example. FIG. **8** illustrates a case where the encryption modules are arranged on both of the upper and lower surfaces of the board assembly **20**, for example.
(71) As illustrated in FIG. **8**, the ventilation holes **123** provided on the housing **10** are arranged also below the lowermost position of the arrangement position range R.sub.0 of the board assembly **20** in the Z-direction, and arranged also above the uppermost position of the same. When the artificial light sources **100** are placed at the ventilation holes **123**, the upper surface or lower surface of the board assembly **20** comes to include a portion to be irradiated with light. In the example of FIG. **8**, the upper surface and lower surface of the board assembly **20** become the second region to be irradiated with light. When the encryption module is present in the second region, the encryption module can be visually observed through the ventilation holes **123**. Further, as the spacers **22**A and **22**B are not provided between the printed circuit boards **21**A to **21**C adjacent in the Z-direction, the encryption module arranged between the printed circuit boards **21**A to **21**C adjacent in the Z-direction ends up being irradiated with light. In other words, the encryption module can be visually observed through the ventilation holes **123**. Accordingly, from the viewpoint to prevent the encryption module from being visually observed, the arrangement of the ventilation holes **123** described above is inapposite.
(72) Here, in the above description, it is designed to prevent the entirety of the encryption module from being visually observed. However, for example, where information to be protected from

visual observation is the production number, model number, design information (such as wire trace and internal structure), or assembling information of the encryption module, it may be designed to allow part of the encryption module, such as its lateral surface, to be visually observed, as long as the information described above cannot be visually observed.

(73) Further, the above description is exemplified by a case where the board assembly **20** accommodated in the housing **10** includes the three printed circuit boards **21**A to **21**C; however, the embodiment is not limited to this. The board assembly **20** may include one or more printed circuit boards.

(74) Further, the above description is exemplified by a case where the spacers **22**A and **22**B respectively fix the printed circuit boards **21**A and **21**B positioned below by the engaging portions **223**; however, the embodiment is not limited to this. The spacers **22**A and **22**B may respectively fix the printed circuit boards **21**B and **21**C positioned above by engaging portions **223**.

(75) Further, the above description is exemplified by a case where the ventilation holes **123** are provided on the side walls **122** of the cover **12**; however, the ventilation holes **123** may be provided on the side walls **112***a* and **112***b* of the base **11**. Further, the ventilation holes **123** are formed in a mesh shape on the side walls **122**; however, the ventilation holes **123** may be formed in another shape, such as a slit or lattice shape.

(76) Further, the above description is exemplified by a case where the ventilation holes **123** are provided on the side walls **122** of the cover **12**; however, the embodiment may be applied to a case where the ventilation holes **123** are provided on the bottom wall **111**, top wall **121**, or another side wall of the housing **10**, for example.

(77) Further, the above description is exemplified by a case where the controller package **23***c* is considered as the encryption module or the controller package **23***c* and nonvolatile memory packages **23***a* are considered as the encryption modules. However, where data to be treated in an encryption process is read into volatile memory packages **23***b* to perform the encryption process, these volatile memory packages **23***b* are also included in the encryption modules. The arrangement positions of the volatile memory packages **23***b* on the upper surface or lower surface of the board assembly **20** may be achieved by the same method as that described above.

(78) In the first embodiment, between the printed circuit boards **21**A to **21**C adjacent in the Z-direction, the spacers **22**A and **22**B are interposed which are formed in a frame shape having almost the same outline as that of the printed circuit boards **21**A to **21**C. In this state, the printed circuit boards **21**A to **21**C and the spacers **22**A and **22**B are fixed to each other to form the board assembly **20**, which is then accommodated in the housing **10** including the side walls **122** formed with the ventilation holes **123**. Here, the controller package **23***c* is present in the first region of the board assembly **20** that is light unreachable when the board assembly **20** is irradiated with light from the artificial light sources **100** placed at the ventilation holes **123**. Consequently, it is possible to prevent the controller package **23***c* from being visually observed from outside the housing **10**.

(79) In other words, it is designed to prevent the production number, model number, design information, such as wire trace and internal connection, or assembling information of the controller package **23***c* in the electronic apparatus **1** from being visually observed from outside the housing **10**. As a result, it is possible to improve the reliability of the electronic apparatus **1** concerning its security.

(80) Further, as the spacers **22**A and **22**B are made of a resin superior in heat radiation property, there is also an effect capable of releasing heat generated in the printed circuit boards **21**A to **21**C adjacent in the Z-direction. Further, the reinforcing parts **222**, **222***a*, and **222***b* of the spacers **22**A and **22**B are provided with the raised portions **224**, and the heat conductive sheets **131** are interposed between the raised portions **224** and the printed circuit boards **21**A to **21**C or devices mounted on the printed circuit boards **21**A to **21**C. Consequently, it is possible to conduct heat from the printed circuit boards **21**A to **21**C to the spacers **22**A and **22**B, and to release the heat by air flows from the ventilation holes **123** provided on the housing **10**.

(81) Further, the spacers **22**A and **22**B are provided with the pins **226** for positioning, and the printed circuit boards **21**A and **21**B are provided with through holes **214** at the positions corresponding to the pins. Consequently, by inserting the pins **226** of the spacers **22**A and **22**B into the through holes **214** of the printed circuit boards **21**A and **21**B, the printed circuit boards **21**A and **21**B can be easily attached to the spacers **22**A and **22**B in a state where the alignment therebetween has been achieved. Further, as the spacers **22**A and **22**B are provided with the engaging portions **223** of a snap fit type, the printed circuit boards **21**A and **21**B can be firmly fixed.

Second Embodiment

(82) In the first embodiment, each spacer arranged between printed circuit boards adjacent in the thickness direction has a structure that surrounds the space between the printed circuit boards, and does not allow air coming from outside to flow into the space between the printed circuit boards. In the second embodiment, an explanation will be given of an electronic apparatus having a structure that allows air from outside to flow also into the space between printed circuit boards adjacent in the thickness direction.

(83) FIGS. **9**A and **9**B are sectional views illustrating an example of an internal configuration of an electronic apparatus according to the second embodiment. FIG. **9**A is a sectional view in a direction perpendicular to the latitudinal direction. FIG. **9**B is a sectional view in a direction perpendicular to the longitudinal direction. FIG. **10** is an exploded perspective view illustrating an example of the electronic apparatus according to the second embodiment. FIG. **11** is an exploded perspective view illustrating an example of a board assembly according to the second embodiment. FIGS. **12**A and **12**B are perspective views illustrating an example of frames according to the second embodiment. FIGS. **13**A to **13**C are top views illustrating examples of a ventilation mechanism of a frame according to the second embodiment. FIG. **14** is a perspective view illustrating another example of the ventilation mechanism of a frame according to the second embodiment. FIGS. **15**A to **15**E are views illustrating an example of an assembling sequence of the board assembly according to the second embodiment. Hereinafter, the electronic apparatus will be exemplified by an SSD, as in the first embodiment. Further, hereinafter, for the sake of convenience, it is assumed that the latitudinal direction of the rectangular top surface or bottom surface of the electronic apparatus **1** is an X-direction, its longitudinal direction is a Y-direction, and its thickness direction is a Z-direction. Further, hereinafter, the relative positional relations of components arranged in the Z-direction, i.e., their relations in the vertical direction, will be illustrated, by using as a reference the arrangement state of the electronic apparatus **1** illustrated in FIGS. **9**A, **9**B, and **10**. Furthermore, hereinafter, an explanation of structures different from those of the first embodiment will be given, while no explanation of structures the same as those of the first embodiment will be given.

(84) As illustrated in FIGS. **9**A, **9**B, and **10**, in the electronic apparatus **1** according to the second embodiment, a housing **10** has the same structure as that of the first embodiment, and a pair of side walls **122** perpendicular to the Y-direction are provided with ventilation holes **123** for cooling, which allow air to flow between the outside and inside of the housing **10**.

(85) In the electronic apparatus **1** according to the second embodiment, a board assembly **20** includes a plurality of printed circuit boards and one or more spacers. The printed circuit boards and the spacers are alternately stacked in the Z-direction, and are fixed by fasteners. In this example, the board assembly **20** has a structure in which three printed circuit boards **21**A, **21**B, and **21**C and two spacers **27**A and **27**B are alternately stacked in the Z-direction. Further, the board assembly **20** includes electrically insulating sheets **29** interposed between the printed circuit boards **21**A and **21**B adjacent in the Z-direction and between the printed circuit boards **21**B and **21**C adjacent in the Z-direction.

(86) Each of the printed circuit boards **21**A to **21**C has a structure the same as that described in the first embodiment. However, in the first embodiment, the printed circuit boards **21**A to **21**C are electrically connected to each other by the board-to-board connectors **24**; on the other hand, in the second embodiment, the printed circuit boards **21**A to **21**C are electrically connected to each other

by flexible printed circuits (Flexible Printed Circuits, each of which will be referred to as “FPC”) **26**A and **26**B.

(87) As illustrated in FIG. **11**, the printed circuit board **21**A is arranged at the middle, the printed circuit board **21**B is arranged at one side (left side in FIG. **11**) of the printed circuit board **21**A in the X-direction, and the printed circuit board **21**C is arranged at the other side (right side in FIG. **11**) of the printed circuit board **21**A in the X-direction. The FPCs **26**A and **26**B each for electrically connecting two printed circuit boards are provided between the printed circuit boards **21**A and **21**B and between the printed circuit boards **21**A and **21**C. Each of the FPCs **26**A and **26**B is composed of a base layer made of a flexible and electrically insulating material and an electrically conductive layer made of an electrically conductive material bonded thereon. As the base layer, polyimide, polyethylene terephthalate, polyethylene naphthalate, or the like is used. As the electrically conductive layer, copper or the like is used. The electrically conductive layer is to be connected to the wiring layer of each printed circuit board.

(88) In this example, the FPCs **26**A and **26**B are arranged at long sides of the printed circuit boards **21**A to **21**C. Further, the lengths of FPCs **26**A and **26**B have been determined in accordance with the stacking order of the printed circuit boards **21**A to **21**C. In this example, as the printed circuit board **21**A, the printed circuit board **21**B, and the printed circuit board **21**C are to be stacked in this order from below, the length of the FPC **26**B that connects the printed circuit boards **21**A and **21**C to each other is set larger than the length of the FPC **26**A that connects the printed circuit boards **21**A and **21**B to each other.

(89) Each of the spacers **27**A and **27**B is interposed between the two printed circuit boards **21**A and **21**B or two printed circuit boards **21**B and **21**C adjacent in the Z-direction. Each of the spacers **27**A and **27**B holds a state where the two printed circuit boards **21**A and **21**B or two printed circuit boards **21**B and **21**C adjacent in the Z-direction have a predetermined interval therebetween. Further, each of the spacers **27**A and **27**B includes a function of making it difficult for devices arranged between the two printed circuit boards **21**A and **21**B or two printed circuit boards **21**B and **21**C adjacent in the Z-direction to be visually observed, through the ventilation holes **123**. Further, each of the spacers **27**A and **27**B includes a function of guiding part of the air flows from the ventilation holes **123** at one side of the housing **10** to the ventilation holes **123** at the other side, through the space between the two printed circuit boards **21**A and **21**B or two printed circuit boards **21**B and **21**C adjacent in the Z-direction.

(90) As illustrated in FIGS. **12**A and **12**B, each of the spacers **27**A and **27**B includes a frame part **271**, which has almost the same outline as that of the printed circuit boards **21**A to **21**C. The frame part **271** is provided with a plurality of engaging portions **273** of a snap fit type at predetermined positions. In the example of FIGS. **12**A and **12**B, the engaging portions **273** are arranged on respective ones of the upper and lower sides of the frame part **271** in the Z-direction. Consequently, each of the spacers **27**A and **27**B comes to be fixed to two of the printed circuit boards **21**A to **21**C, which are arranged above and below this one of the spacers **27**A and **27**B in the Z-direction. Here, when the spacers **27**A and **27**B are seen in a state placed by the same orientation, the spacer **27**A is provided with the engaging portions **273** on a long side at one side, and the spacer **27**B is provided with the engaging portions **273** on a long side at the other side. The engaging portions **273** are preferably arranged at positions that prevent the printed circuit boards **21**B and **21**C from being opened up by a reaction force of the FPCs **26**A and **26**B when the board assembly **20** is assembled. Further, as illustrated in FIG. **15**A, in accordance with the positions of the engaging portions **273**, the printed circuit boards **21**B and **21**C are provided with recessed portions **217** on long sides opposite to the long sides provided with the FPCs **26**A and **26**B. When the printed circuit boards **21**B and **21**C are fixed to the spacers **27**A and **27**B, the engaging portions **273** engage with the recessed portions **217**; thus, the printed circuit boards **21**B and **21**C are fixed, and are prevented from being opened up by a reaction force of the FPCs **26**A and **26**B.

(91) The frame part **271** is provided with through holes **275** for inserting the screws **142** that fix the

board assembly **20** to the base **11**. Further, the frame part **271** is provided with a plurality of pins **276** projected downward and upward. The pins **276** are used to perform positioning in the X-direction and Y-direction with respect to each of the printed circuit boards **21**A and **21**B to be fitted. Accordingly, the printed circuit boards **21**A to **21**C are provided with through holes **214** for fitting the pins **276** therein, at the corresponding positions.

(92) As illustrated in FIG. **11**, in assembling, first, the spacers **27**A and **27**B are respectively put closer to the upper sides of the printed circuit boards **21**A and **21**B. Then, while the positions of the pins **276** of the spacers **27**A and **27**B are set aligned with the positions of the through holes **214** of the printed circuit boards **21**A and **21**B, the spacers **27**A and **27**B are respectively brought into contact with the printed circuit boards **21**A and **21**B. This results in a state where the protrusions of the engaging portions **273** are positioned on the lower surface sides of the printed circuit boards **21**A and **21**B, and engage with these lower surfaces. Further, the printed circuit boards **21**B and **21**C are respectively put closer to the upper sides of the spacers **27**A and **27**B. Then, while the positions of the through holes **214** of the printed circuit boards **21**B and **21**C are set aligned with the positions of the pins **276** of the spacers **27**A and **27**B, the printed circuit boards **21**B and **21**C are respectively brought into contact with the spacers **27**A and **27**B. This results in a state where the protrusions of the engaging portions **273** are positioned on the upper surface sides of the printed circuit boards **21**B and **21**C, and engage with these upper surfaces. In this way, the printed circuit boards **21**A to **21**C are fixed to the spacers **27**A and **27**B.

(93) As illustrated in FIGS. **12**A and **12**B, the side surface **271***a* of the frame part **271** to be opposite to the ventilation holes **123** is provided with a ventilation mechanism. The ventilation mechanism is configured such that slits are formed in the side surface **271***a* of the frame part **271**. As illustrated in FIG. **12**A, two types of slits **2711** and **2712** different in direction are alternately formed in the side surface **271***a* along the X-direction. As a result, wall portions **2715***a* and **2715***b*, each of which has a pentagonal shape with an apex protrusive in the Y-direction, are formed at places surrounded by the two types of slits **2711** and **2712**. The pentagonal wall portions **2715***a* and **2715***b* adjacent in the X-direction have different apex directions, that is, one of them is toward a positive direction side in the Y-direction and the other is toward a negative direction side in the Y-direction.

(94) Further, as illustrated in FIGS. **9**A and **11**, the spacer **27**A is arranged on the lower side in the Z-direction. Further, the printed circuit board **21**A is arranged below the spacer **27**A in the Z-direction, and capacitors **23***d* are arranged at one end of the printed circuit board **21**A in the Y-direction. The capacitors **23***d* are present outside a side surface **271***b* of the frame part **271**. The two capacitors **23***d* present in the X-direction serve to block the field of vision into the board assembly **20**. Accordingly, as illustrated in FIG. **12**A, the side surface **271***b* of spacer **27**A is structured such that no side surface portion is formed at the positions where the capacitors **23***d* are to be arranged, but one pentagonal wall portion **2715***a* is formed to be present between the two capacitors **23***d*.

(95) On the other hand, in the spacer **27**B arranged on the upper side in the Z-direction, as illustrated in FIG. **12**B, the side surfaces **271***a* and **271***b* to be opposite to the ventilation holes **123** are provided with pentagonal wall portions **2715***a* and **2715***b*.

(96) As illustrated in FIGS. **13**A to **13**C, it is assumed that, when the side surfaces **271***a* and **271***b* are seen in the Y-direction, the distance between the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b* is denoted by “d”. The distance between the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b* is preferably set to be zero as illustrated in FIG. **13**A, i.e., there should be no gap between the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b*. Alternatively, the distance between the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b* is preferably set to be negative, i.e., there should be a partial overlap near the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b*. With this arrangement, as described in the first embodiment, it is possible to prevent the encryption module in the board assembly **20** from being visually observed through the ventilation holes **123** of

the housing **10**. However, as the ventilation holes **123** provided on the side walls **122** of the housing **10** are formed in a mesh shape, the net portions between the ventilation holes **123** overlap with the slits of the side surface **271***a* of the frame part **271**, and make it difficult for the encryption module inside to be visually observed. Accordingly, as illustrated in FIG. **13**C, it may be designed to leave a gap of about 0.1 to 0.2 mm between the mutually adjacent pentagonal wall portions **2715***a* and **2715***b*. If the distance “d” between the mutually adjacent ends of the pentagonal wall portions **2715***a* and **2715***b* is larger than 0.2 mm, it may become possible for the encryption module inside to be visually observed. Thus, the distance “d” is preferably set to be up to 0.2 mm.
(97) Here, the slits **2711** and **2712** provided on the spacers **27**A and **27**B are a mere example, and another structure may be provided instead. For example, as illustrated in FIG. **14**, slits **2712** may be formed in parallel with each other on the side surface **271***a* or side surface **271***b* of the frame part **271** along the X-direction. In this case, wall portions **2715***c*, each of which has a parallelogram shape, are formed at places surrounded by respective sets of mutually adjacent two slits **2712**. The distance between the mutually adjacent ends of the parallelogram wall portions **2715***c* also satisfies the relation described with reference to FIGS. **13**A to **13**C. Other than this, a slit structure which allows air to flow through the inside of the board assembly **20** and makes it difficult for the inside to be visually observed may be used. Further, for example, the spacers **27**A and **27**B are made of polycarbonate resin.
(98) The electrically insulating sheets **29** are interposed between the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction. The electrically insulating sheets **29** serve to prevent devices on the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction from being damaged by coming into contact with each other, and to relax impact shock when the electronic apparatus **1** receives the impact shock. Further, each electrically insulating sheet **29** includes a function of electrically insulating the portion between two of the printed circuit boards **21**A to **21**C mutually adjacent.
(99) Next, with reference to FIGS. **15**A to **15**E, an explanation of a method of assembling such the electronic apparatus **1** will be given. However, since a method of fixing the board assembly **20** to the housing **10** to assemble the electronic apparatus **1** is the same as that described in the first embodiment, its description will be omitted; thus, a method of assembling the board assembly **20** by using the printed circuit boards **21**A to **21**C, the spacers **27**A and **27**B, and the electrically insulating sheets **29** will be described.
(100) Here, as illustrated in FIG. **15**A, a group **21** of the printed circuit boards connected by the FPCs **26**A and **26**B is placed in an unfolded state. In this example, in a state where the connector **25** of the printed circuit board **21**A is present on the upper side in FIG. **15**A, the printed circuit board **21**B is present at the left side of the printed circuit board **21**A next to the FPC **26**A, and the printed circuit board **21**C is present at the right side of the printed circuit board **21**A next to the FPC **26**B.
(101) Then, as illustrated in FIG. **15**B, while the pins (not illustrated) on the lower side of the spacer **27**A are set aligned with the through holes **214** of the printed circuit board **21**A at the middle, the spacer **27**A is put closer to the printed circuit board **21**A from above in the Z-direction. Then, the spacer **27**A is fixed to the printed circuit board **21**A by the engaging portion **273** on the lower side of the spacer **27**A.
(102) Thereafter, the electrically insulating sheet **29** (not illustrated) is placed on the printed circuit board **21**A with the spacer **27**A attached thereon. Then, as illustrated in FIG. **15**C, the printed circuit board **21**B is folded back above the printed circuit board **21**A. At this time, while the through holes **214** of the printed circuit board **21**B are set aligned with the pins **276** on the upper side of the spacer **27**A, the printed circuit board **21**B is put closer to the spacer **27**A from above in the Z-direction. Then, the printed circuit board **21**B is fixed to the spacer **27**A by the engaging portion **273** on the upper side of the spacer **27**A.
(103) Then, as illustrated in FIG. **15**D, while the pins (not illustrated) on the lower side of the spacer **27**B are set aligned with the through holes **214** of the printed circuit board **21**B, the spacer

27B is put closer to the printed circuit board **21**B from above in the Z-direction. Then, the spacer **27**B is fixed to the printed circuit board **21**B by the engaging portion **273** on the lower side of the spacer **27**B.
(104) Thereafter, the electrically insulating sheet **29** (not illustrated) is placed on the printed circuit board **21**B with the spacer **27**B attached thereon. Then, as illustrated in FIG. **15**E, the printed circuit board **21**C is folded back above the printed circuit board **21**B. At this time, while the through holes **214** of the printed circuit board **21**C are set aligned with the pins **276** on the upper side of the spacer **27**B, the printed circuit board **21**C is put closer to the spacer **27**B from above in the Z-direction. Then, the printed circuit board **21**C is fixed to the spacer **27**B by the engaging portion **273** on the upper side of the spacer **27**B. As a result, the board assembly **20** is constructed.
(105) Here, in the second embodiment, the board assembly **20** is assembled by using the group **21** of the printed circuit boards in which the printed circuit boards **21**A to **21**C are mutually connected by the FPCs **26**A and **26**B. However, the spacers **27**A and **27**B including the slits described above may be used for a board assembly **20** having a structure in which the printed circuit boards **21**A to **21**C are connected to each other by the board-to-board connectors **24** described in the first embodiment. Further, also in the second embodiment, the relation between the positions of the ventilation holes **123** of the housing **10** and the arrangement position of the encryption module in the board assembly **20** satisfies the relation described in the first embodiment.
(106) In the second embodiment, between the printed circuit boards **21**A to **21**C adjacent in the Z-direction, the spacers **27**A and **27**B are interposed which have almost the same outline as that of the printed circuit boards **21**A to **21**C. The spacers **27**A and **27**B include the side surfaces **271***a* and **271***b* formed with the slits and being opposite to the ventilation holes **123** of the housing **10**. In this state, the printed circuit boards **21**A to **21**C and the spacers **27**A and **27**B are set to form the board assembly **20**, which is then accommodated in the housing **10**. Consequently, air flows from the ventilation holes **123** of the housing **10**, through the slits on one side of the spacers **27**A and **27**B, also into the board assembly **20**, and is exhausted from the slits on the other side. Thus, there is an effect capable of improving the cooling effect on devices that generate heat inside the board assembly **20**. Particularly, there is a case where the controller package **23***c*, which is formed of the SoC and is large in heat generation, is attached to the lower surface, in the Z-direction, of the printed circuit board **21**A. In this case, heat is conducted also to the upper surface, in the Z-direction, of the printed circuit board **21**A, at the position where the controller package **23***c* is attached. Accordingly, when air flows along the Y-direction through the slits of the spacer **27**A, the back side at the position where the controller package **23***c* is attached is cooled by air.
(107) Further, when the side walls **122** of the housing **10** provided with the ventilation holes **123** are seen in the Y-direction, the net portions of the housing **10** overlap with the slits of the spacers **27**A and **27**B. This makes it difficult for the inside of the board assembly **20** to be visually observed through the slits. Further, when the ends of wall portions of the spacers **27**A and **27**B provided at positions opposite to the ventilation holes **123** are set to overlap with each other, the inside of the board assembly **20** can be prevented from being visually observed through the slits, when seen in the Y-direction. Consequently, it is possible to improve the reliability of the electronic apparatus **1** concerning its security.
Third Embodiment
(108) In the first and second embodiments, the ventilation hole provided on the housing are formed in a mesh shape, and the side surfaces or the like of the board assembly, as long as no encryption module is on them, can be visually observed through the ventilation holes. In the third embodiment, an explanation of an electronic apparatus that does not allow the inside of the housing to be visually observed through the ventilation holes provided on the housing will be given.
(109) FIGS. **16**A and **16**B are views illustrating an example of the cover of a housing according to a third embodiment. FIG. **16**A is a perspective view, and FIG. **16**B is a sectional view. FIG. **17** is a sectional view illustrating another example of the cover of the housing according to the third

embodiment. FIG. **18** is a perspective view illustrating an example of a board assembly according to the third embodiment. FIG. **19** is a perspective view illustrating a configuration example of a spacer according to the third embodiment.

(110) In the third embodiment, as illustrated in FIGS. **16**A and **16**B, each ventilation hole **123** formed on the side walls **122** of the cover **12** is provided with a recessed portion **1221** formed by louver working. The recessed portion **1221** is formed with its lower end and both ends in the X-direction continued to the surface of the corresponding side wall **122**, and formed with an inclined surface bent toward the inside of the housing **10** and extending upward within the range of the corresponding ventilation hole **123**. The position of the upper end of the recessed portion **1221** is almost the same as the position of the upper end of the corresponding ventilation hole **123**. In other words, the recessed portion **1221** is arranged to substantially cover the corresponding ventilation hole **123** from visual observation on the housing **10** in the Y-direction. Accordingly, when the housing **10** is visually observed in the Y-direction, the inside of the housing **10** is prevented from being visually observed. Here, FIGS. **16**A and **16**B illustrate a structure in which the recessed portion **1221** is formed on the lower end side of the corresponding ventilation hole **123**, and causes air to flow upward. However, this is a mere example, and the recessed portion **1221** may be formed at the corresponding ventilation hole **123** to cause air to flow sideward or downward.

(111) Further, as illustrated in FIG. **17**, each ventilation hole **123** formed on the side walls **122** of the cover **12** may be provided with a cut-and-raised portion **1222** formed by press working. The cut-and-raised portion **1222** is formed with its lower end continued to the surface of the corresponding side wall **122**, and formed with an inclined surface present on the inner side of the housing **10** and extending upward within the range of the corresponding ventilation hole **123**. The position of the upper end of the cut-and-raised portion **1222** is almost the same as the position of the upper end of the corresponding ventilation hole **123**. In other words, the cut-and-raised portion **1222** is arranged so that it can cover the corresponding ventilation hole **123** when the housing **10** is visually observed in the Y-direction. Here, FIG. **17** illustrates a structure in which the cut-and-raised portion **1222** is formed on the lower end side of each ventilation hole **123**, and causes air to flow upward. However, this is a mere example, and the cut-and-raised portion **1222** may be formed at each ventilation hole **123** to cause air to flow sideward or downward.

(112) As described above, each ventilation hole **123** on the side walls **122** of the housing **10** is provided with a blindfold, such as the recessed portion **1221** or cut-and-raised portion **1222**, not to allow the inside of the housing **10** to be visually observed. Consequently, it is possible to improve the reliability of the electronic apparatus **1** concerning its security. Here, as a spacer used for the board assembly **20** according to the third embodiment, either one of the spacers described in the first and second embodiments can be used; furthermore, a spacer which does not have a structure described in the first and second embodiments, which prevents the encryption module from being visually observed from outside, may be used. As illustrated in FIG. **18**, it is sufficient if spacers **28**A and **28**B have a structure that can hold a state where the printed circuit boards **21**A to **21**C adjacent in the Z-direction have predetermined intervals therebetween.

(113) Each of the spacers **28**A and **28**B has almost the same dimensions as those of the printed circuit boards **21**A to **21**C, and has almost the same outline as that of the printed circuit boards **21**A to **21**C, which is substantially rectangular. As illustrated in FIG. **19**, each of the spacers **28**A and **28**B includes column members **281** arranged near the four corners to maintain, at a predetermined interval therebetween, two of the printed circuit boards **21**A to **21**C mutually adjacent in the Z-direction, and beam members **282** connecting the column members **281** to each other. In the third embodiment, the thickness of each beam member **282** in the Z-direction is set smaller than that of each column member **281**. Accordingly, when the board assembly **20** has been assembled, gaps are formed as passages of air flow between the printed circuit boards **21**A to **21**C adjacent in the Z-direction.

(114) The beam members **282** are provided with a plurality of engaging portions **283** of a snap fit

type at predetermined positions. In the example of FIG. **19**, the engaging portions **283** are arranged on respective ones of the upper and lower sides in the Z-direction. Consequently, each of the spacers **28**A and **28**B comes to be fixed to two of the printed circuit boards **21**A to **21**C, which are arranged above and below this one of the spacers **28**A and **28**B in the Z-direction. Here, as in the second embodiment, it may be designed such that, when the spacers **28**A and **28**B are seen in a state placed by the same orientation, the spacer **28**A is provided with the engaging portions **283** on a long side at one side, and the spacer **28**B is provided with the engaging portions **283** on a long side at the other side. In a case where the FPCs **26**A and **26**B form the connection between the printed circuit boards **21**A and **21**B and the connection between the printed circuit boards **21**B and **21**C, as in the second embodiment, the engaging portions **283** are preferably arranged at positions that prevent the printed circuit boards **21**B and **21**C from being opened up by a reaction force of the FPCs **26**A and **26**B when the board assembly **20** is assembled. Further, as illustrated in FIG. **15**A, in accordance with the positions of the engaging portions **283**, the printed circuit boards **21**B and **21**C are provided with recessed portions **217** on long sides opposite to the long sides provided with the FPCs **26**A and **26**B. When the printed circuit boards **21**B and **21**C are fixed to the spacers **28**A and **28**B, the engaging portions **283** engage with the recessed portions **217**; thus, the printed circuit boards **21**B and **21**C are fixed, and are prevented from being opened up by a reaction force of the FPCs **26**A and **26**B.

(115) The column members **281** are provided with through holes **285** for inserting the screws **142** that fix board assembly **20** to the base **11**. Further, the column members **281** are provided with a plurality of pins **286** projected downward and upward. The pins **286** are used to perform positioning in the X-direction and Y-direction with respect to each of the printed circuit boards **21**A to **21**C to be fitted. Accordingly, the printed circuit boards **21**A to **21**C are provided with through holes **214** for fitting the pins **286** therein, at the corresponding positions.

(116) Here, the constituent elements corresponding to those of the first and second embodiments are denoted by the same reference symbols, and their description will be omitted. Further, as a method of assembling the electronic apparatus **1** according to the third embodiment, the same method as that described in the first or second embodiment may be used. Further, also in the third embodiment, the arrangement position of the encryption module in the board assembly **20** is set at a position that cannot be visually observed through the ventilation holes **123** of the housing **10**, as in the first embodiment.

(117) In the third embodiment, each ventilation hole **123** of the housing **10** is provided with a blindfold. Consequently, even though the ventilation mechanism is provided, it is possible to prevent the inside of the housing **10** from being visually observed. Further, as a structure of the housing **10** is used to prevent the inside of the housing **10** from being visually observed, it is possible to determine the arrangement position of the encryption module in the board assembly **20** inside the housing **10** and the arrangement positions of the ventilation holes **123**, without suffering from the restriction described in the first embodiment. Further, for the board assembly **20**, a spacer can be used which does not have a structure for preventing an encryption module from being visually observed from outside. In this case, it is possible to further improve flows of air into the board assembly **20**.

(118) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A memory system comprising: a first printed circuit board having a first main surface and a second main surface, the second main surface opposing the first main surface; a nonvolatile memory; a first encryption device mounted on the first main surface, the first encryption device being configured to encrypt data to be stored in the nonvolatile memory; and a housing including a wall and a first plate, the wall being perpendicular to the first plate, the first plate being parallel to the first printed circuit board, the housing accommodating the first printed circuit board, the nonvolatile memory, and the first encryption device, the wall including a first region and a second region, the first region including an opening, the second region including no opening, the first plate having a first surface, the first surface facing the first main surface of the first printed circuit board, wherein the second region of the wall is bounded by a first position and a second position, the first position being a position where the wall is intersected by an imaginary extension of the first main surface of the first printed circuit board, and the second position being a position where the wall is intersected by an imaginary extension of the first surface of the first plate.
2. The memory system according to claim 1, wherein the first encryption device is not irradiated with light emitted from a light source if the light source is placed at the opening.
3. The memory system according to claim 2, further comprising: a second printed circuit board having a third main surface and a fourth main surface, the fourth main surface opposing the third main surface, the third main surface facing the second main surface of the first printed circuit board; and a second encryption device mounted on the fourth main surface, the second encryption device being configured to encrypt data to be stored in the nonvolatile memory, wherein the housing further accommodates the second printed circuit board and the second encryption device, the housing further includes a second plate having a second surface, the second surface facing the fourth main surface of the second printed circuit board, the wall further includes a third region, the third region including no opening, wherein the third region is bounded by a third position and a fourth position, the third position being a position where the wall is intersected by an imaginary extension of the fourth main surface of the second printed circuit board, the fourth position being a position where the wall is intersected by an imaginary extension of the second surface of the second plate, and the second encryption device is not irradiated with light emitted from the light source if the light source is placed at the opening.
4. The memory system according to claim 3, further comprising: a third encryption device mounted on the second main surface of the first printed circuit board, the third encryption device being configured to encrypt data to be stored in the nonvolatile memory; and a spacer interposed between the first printed circuit board and the second printed circuit board, a part of the spacer being located between the opening and the third encryption device, wherein a side of the part of the spacer facing the opening is irradiated with light emitted from the light source if the light source is placed at the opening, and the third encryption device is not irradiated with light emitted from the light source if the light source is placed at the opening.
5. The memory system according to claim 4, wherein the spacer includes a slit configured to allow air to flow from the opening into a region that is surrounded by the first printed circuit board, the second printed circuit board, and the spacer.
6. The memory system according to claim 4, wherein the wall is further perpendicular to the second plate.
7. The memory system according to claim 4, wherein each of the first encryption device, the second encryption device, and the third encryption device includes one of a controller package, a volatile memory package, and a nonvolatile memory package, the controller package accommodating a controller chip configured to control the nonvolatile memory, the volatile memory package accommodating a volatile memory chip, the nonvolatile memory package

accommodating a nonvolatile memory chip.
8. The memory system according to claim 1, wherein the housing has a rectangular parallelepiped shape.
9. The memory system according to claim 1, wherein the first encryption device and the first surface of the first plate are respectively in contact with both sides of a heat conductive member that is placed between the first encryption device and the first plate.
10. The memory system according to claim 1, wherein the first encryption device includes a controller package, the controller package accommodating a controller chip configured to control the nonvolatile memory.
11. A memory system comprising: a first printed circuit board having a first main surface and a second main surface, the second main surface opposing the first main surface; a second printed circuit board having a third main surface and a fourth main surface, the fourth main surface opposing the third main surface, the third main surface facing the second main surface of the first printed circuit board; a spacer interposed between the first printed circuit board and the second printed circuit board, the spacer including a first part and a second part, a gap between the first part and the second part forming a first slit; a nonvolatile memory; an encryption device mounted on the second main surface of the first printed circuit board, the encryption device being configured to encrypt data to be stored in the nonvolatile memory; and a housing including a first wall, the first wall being perpendicular to the first printed circuit board and the second printed circuit board, the housing accommodating the first printed circuit board, the second printed circuit board, the spacer, the nonvolatile memory, and the encryption device, the first wall including at least one first opening, wherein the first part of the spacer and the second part of the spacer locate between the first opening and the encryption device, and the first part of the spacer or the second part of the spacer is placed such that the first part of the spacer or the second part of the spacer blocks light emitted from a light source not to irradiate the encryption device if the light source is placed at the first opening.
12. The memory system according to claim 11, wherein the first slit of the spacer is configured to allow air to flow from the first opening into a region that is surrounded by the first printed circuit board, the second printed circuit board, and the spacer.
13. The memory system according to claim 12, wherein the housing further includes a second wall facing the first wall, the second wall including a second opening, the spacer further includes a third part and a fourth part respectively facing the first part and the second part, a gap between the third part and the fourth part forming a second slit, the third part and the fourth part being located between the second opening and the encryption device, and the second slit of the spacer is configured to allow air flowed from the first opening into the region to flow out through the second opening.
14. The memory system according to claim 11, wherein the at least one first opening comprises a plurality of first openings, the first wall includes a mesh-shaped portion, the mesh-shaped portion forming the plurality of first openings, and a net portion of the mesh-shaped portion between two adjacent first openings overlaps the first slit of the spacer when seen in a direction perpendicular to the first wall.
15. The memory system according to claim 11, wherein the first part of the spacer and the second part of the spacer are formed with no distance therebetween when seen in a direction perpendicular to the first wall.
16. The memory system according to claim 11, wherein the first part of the spacer and the second part of the spacer are formed with a distance therebetween when seen in a direction perpendicular to the first wall.
17. The memory system according to claim 11, wherein at least one of the first part of the spacer and the second part of the spacer has a polygonal shape when seen in a first direction, the first direction being parallel to the first wall.

18. The memory system according to claim 17, wherein an apex having the largest angle among angles of the polygonal-shaped first part of the spacer protrudes in a second direction, the second direction being perpendicular to the first wall, and an apex having the largest angle among angles of the polygonal-shaped second part of the spacer protrudes in a third direction, the third direction being perpendicular to the first wall and opposite to the second direction.
19. The memory system according to claim 17, wherein the polygonal shape is a pentagon.
20. The memory system according to claim 11, wherein the encryption device includes one of a controller package, a volatile memory package, and a nonvolatile memory package, the controller package accommodating a controller chip configured to control the nonvolatile memory, the volatile memory package accommodating a volatile memory chip, the nonvolatile memory package accommodating a nonvolatile memory chip.